

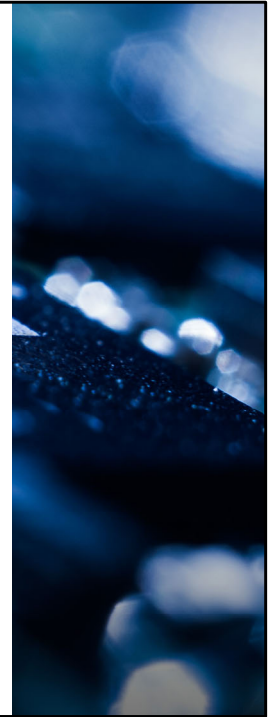


SC1006

Computer Organisation and Architecture

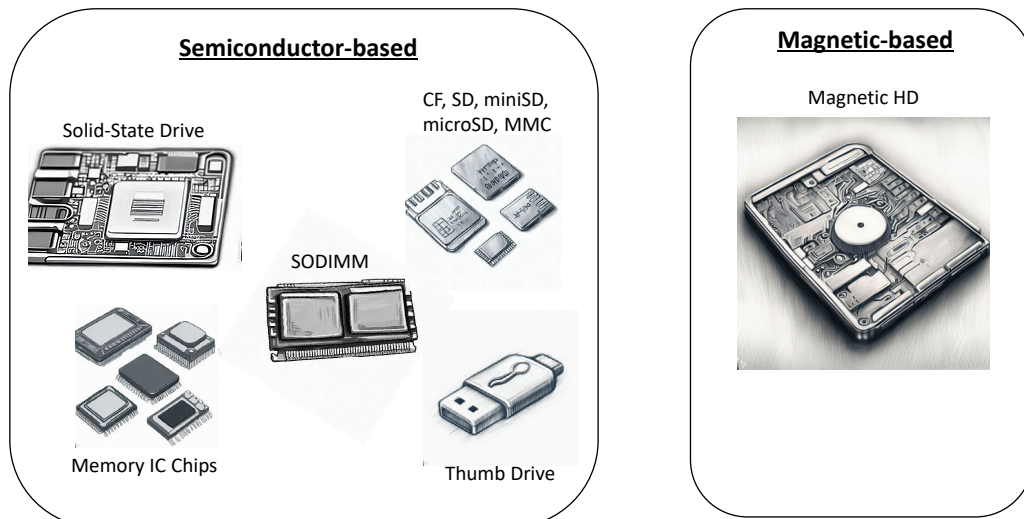
Computer Memory Introduction

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- This section is an introduction on Computer Memory

Computer Memory



Computer memory is a key component in a computer system as this is where all the information and code are stored.

There are many types of memory:

Semiconductor based

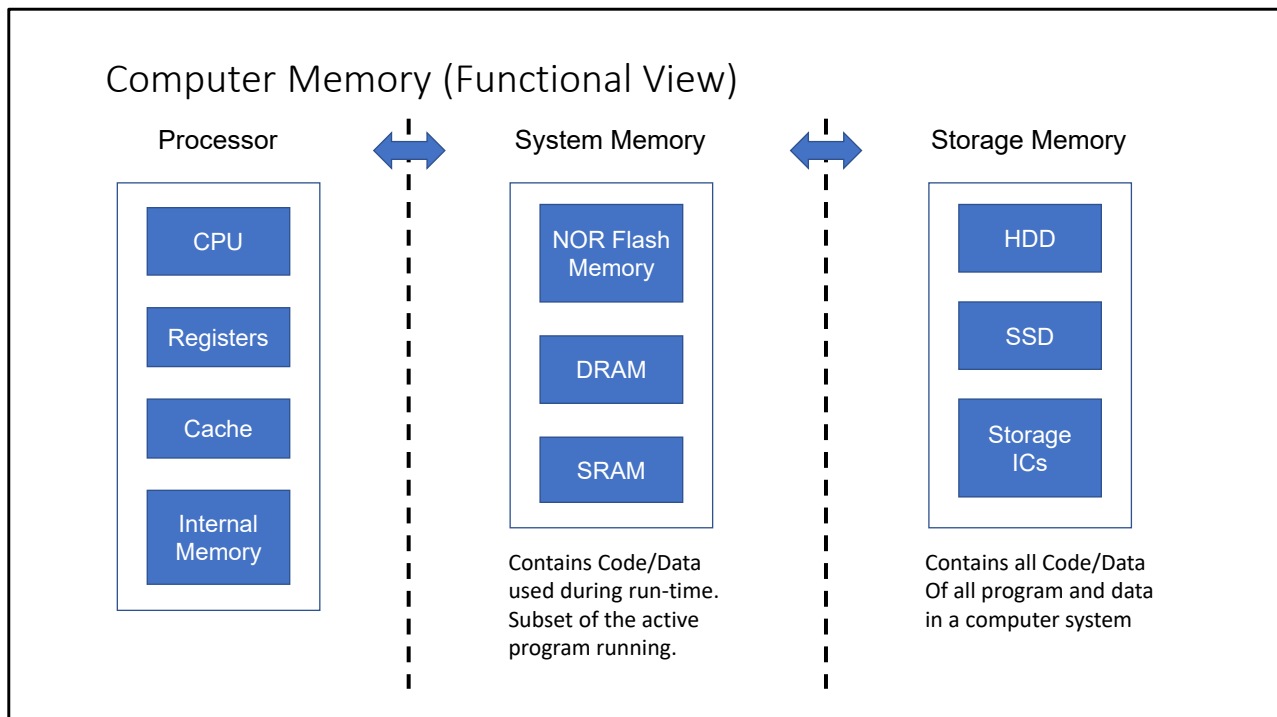
And the Good old Magnetic HDD

Although HDD is fast being replaced by SSD (solid state drive) in laptops these days, you may want to know that HDD is still very much involved in your daily life, in the form of cloud storage.

The iCloud, Google Drive, and drop box cloud storage that you use store information in Data Centers, and HDD is still the main storage element in Data Centers. We will get to more of that in later slides.

One question, among these memories, which are volatile, and which are non-volatile?

To know the answer, we must first understand what is volatile and non-volatile memory.



Before we go into the memory type details, let's go through some basics on the function of a memory in a computer system.

When we look at memory from functional point of view, there are two types of memory in a computer system.

System and Storage memory

System memory is where you run the code directly from. But it is most likely that system memory will not be large enough to contain all the application program and data you have, as such you will need a larger space for all information.

This is where Storage memory comes in. The entire collection of Program and Data is stored in the Storage memory.

Another common attribute of storage memory is that it is typically non-volatile as it needs to preserve the program and data even after the computer power down.

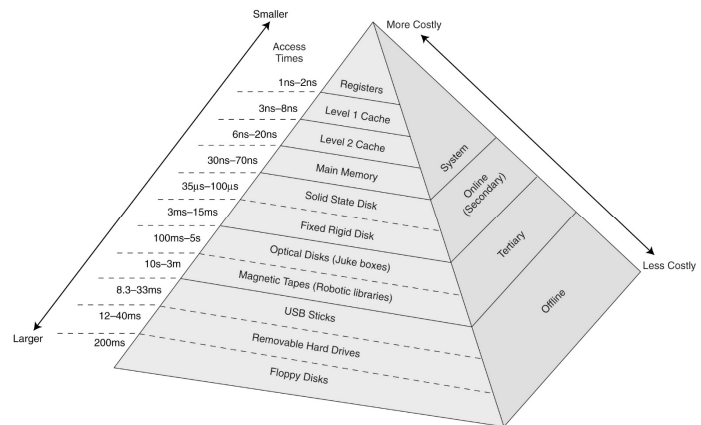
For this course, where system memory is concerned, we only have three memory types to select from: NOR Flash, DRAM and SRAM; the primary reason behind is that these three memories support XIP (**execute in place**), meaning you can run code directly from these memory. No worries if you are not familiar with these names and terms, we will get to more details in later slides.

The processor itself will also have its own internal memory and other memory elements such as Cache and Registers.

Operation wise, the computer will power up with its OS and applications resided in the Storage memory, these are then transferred over to the System memory so that the processor can execute the code.

Memory – Cost vs Function Trade Off

- The memory and storage devices may be organized like a pyramid.
- The pinnacle has the fastest access time but is also more costly.
- Memory Access Time below is only for illustration. These changes with improvement in technology.

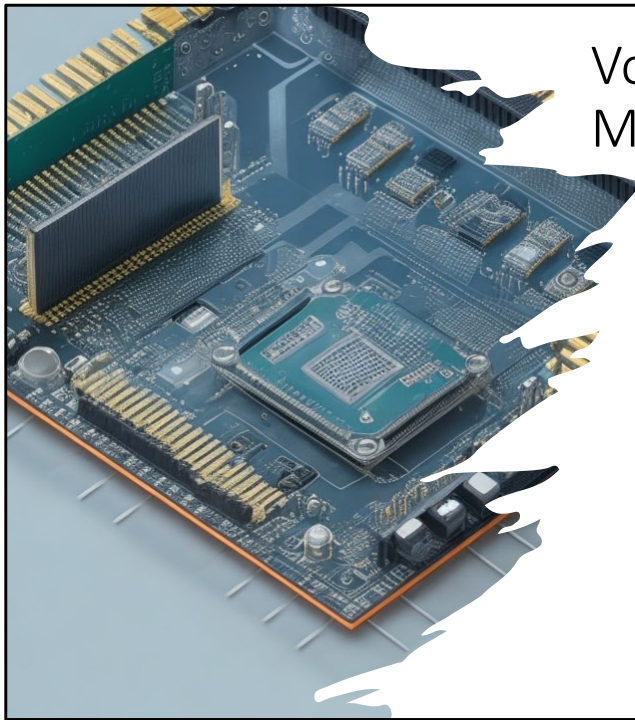


When choosing which type of memory to use, we typically must consider the cost vs function trade off.

IN this slide, different memory types are organised in the form of a pyramid.

The fastest memories are at the top but they are also the most expensive, so their size are typically smaller as compared to those at the bottom.

Note that the memory access timing are for illustration only, these changes with improvement in technology.



Volatile and Non-Volatile Memory

- Volatile
 - Data is lost when electric power is removed.
 - Temporary storage.
 - Typically used as system memory.
 - We will look at Random Access Memories such as Static-RAM (SRAM) and Dynamic-RAM (DRAM).
- Non-volatile
 - Data is retained even if electric power is removed.
 - Permanent storage.
 - Typically used as main storage.
 - We will look at FLASH, magnetic hard-disk specifically.

Back to our discussion on Volatile and Non-Volatile memory
Volatile memories has the attribute that

The data will be lost when power supply to the memory is cut off.

So the information can only last if there is power and is therefore used as a temporary storage.

System memory are typically implemented using volatile memories. For this course, we will look at SRAM and DRAM

Non-volatile memory, on the other hand, is able to preserve the data even when the power is cut.

So it is used as a permanent storage. We will look at Flash and Magnetic Hard Disk in our course.

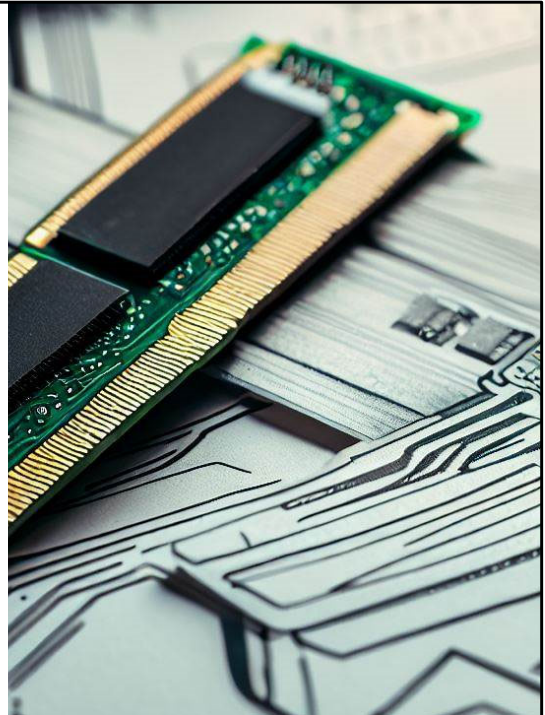
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Semiconductor Memories

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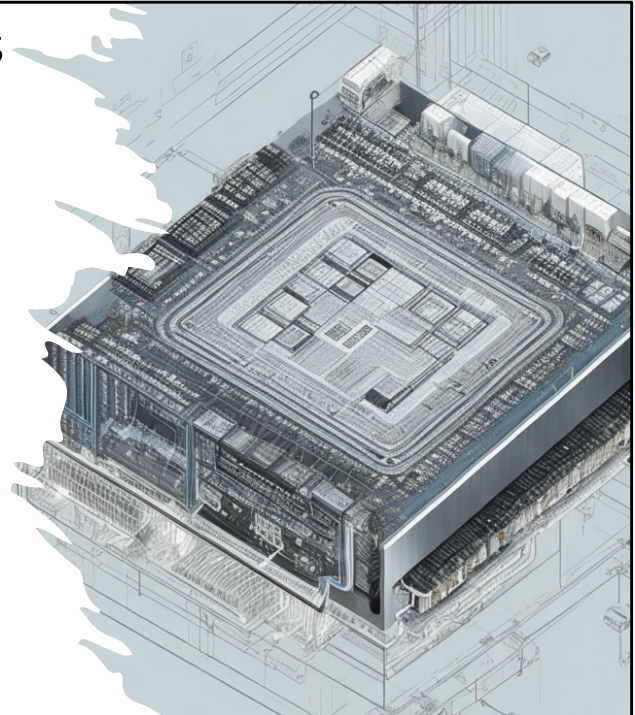
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- The first group of memories we are going to check out is the Semiconductor memories.

Semiconductor Memories

- Memories based on semiconductor integrated circuits (IC).
- Used as processor internal memories and system memory in a computer system
- Processor internal memories
 - Registers
 - Buffers
 - Cache
 - Internal System and Storage Memory (SRAM, DRAM, Flash based)
- Types of memory
 - SRAM
 - DRAM
 - Flash Memory
 - Solid State Drives (based on Flash Memories)



- Semiconductor memories are based on semiconductor IC, short for integrated circuits.
- They are used as processor internal memory elements such as registers, buffers, cache and internal system or storage memory.
- Types of semiconductor memories we will touch on in this course are SRAM, DRAM, Flash Memory and SSD (bulk storage memory (in an array) mainly consists of Flash memory ICs).
- We will discuss more of these in video that follows.

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